Into the Matrix

Exploration of Molecular Symmetry and Bonding Using Jupyter Notebooks

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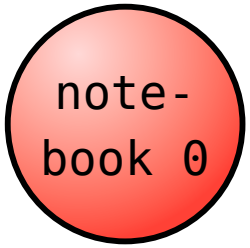
Context

- The Catholic University of America
  - 3,200 undergrads
  - 2,000 grads
  - Recently designated R1
- Chemistry @ CUA
  - (Bio)Chemistry major/minor
  - ~8 graduates/year

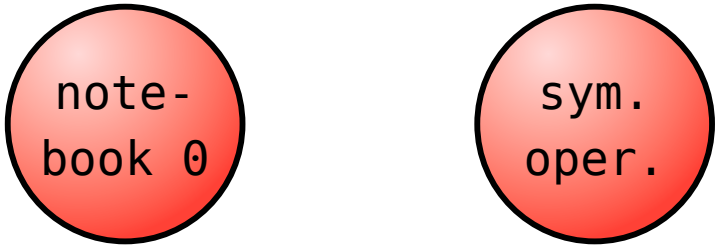
Outcomes

- By the end of this module, you will be able to:
  - use py3Dmol to represent molecules in 3D space.
  - represent symmetry operations in Cartesian coordinates as matrix multiplications.
  - enumerate the combinations of symmetry operators to form a point group and build a character table.
  - apply the character tables to the creation of molecular orbital (MO) diagrams.

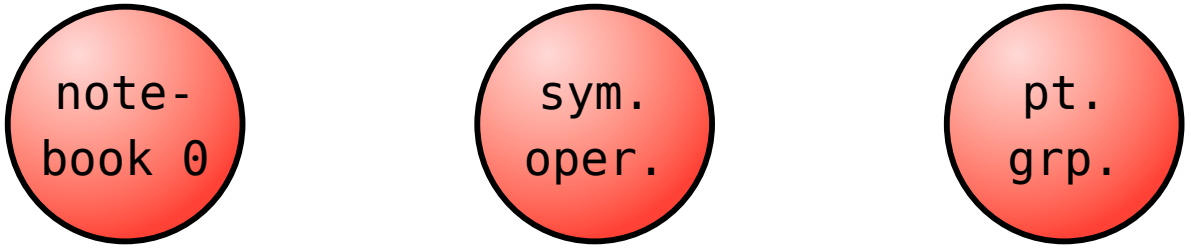
Breakdown



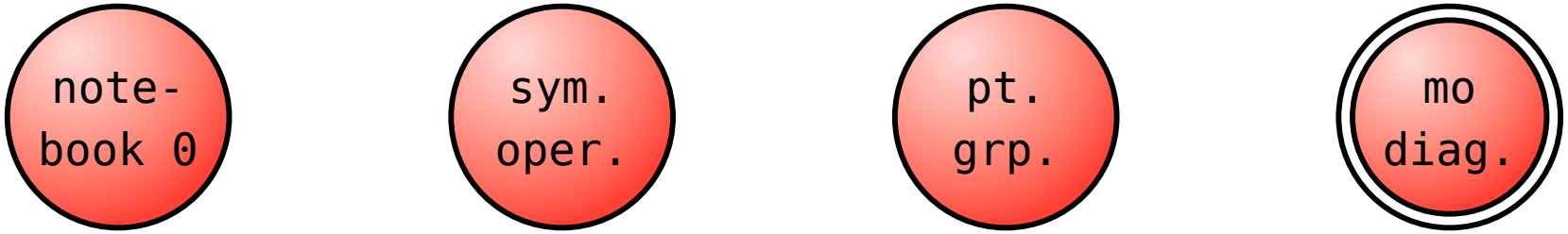
Breakdown



Breakdown



Breakdown



Sample

```
import numpy as np

water = np.array([
   ['atom', 'x',  'y',  'z'],
   ['O',     0,    0,     0],
   ['H',     0,    1,     1],
   ['H',     0,   -1,    -1],
])
```

Sample cont...

```
rot_mat = np.array([
    [1,  0,  0],
    [0,  0,  1],
    [0, -1,  0],
])

water_xyz = np.array([x[1:] for x in
water[1:]]).astype(int)
```

Sample cont...

```
print(water)
print(water_xyz)
print(rot_mat)

water_xyz_rot = np.matmul(water_xyz, rot_mat)
print(water_xyz_rot)
```

Thank You

Questions?